An Exact Local Extension and Homothetic Classes for OEIS A230543

Carlo Corti
Independent researcher
carlo.corti@outlook.com

October 2026

Abstract

OEIS A230543 lists the positive integers whose first two arithmetic derivatives, together with the original integer, form the ordered legs of an integer Pythagorean quadruple. Starting from an accepted 22-term baseline, a complete C enumeration through 130000000000 gives 18 further terms and a 40-term local prefix. This indexing relies on the accepted baseline and the program's negative coverage; a second independent enumeration of all negatives was not performed. All 40 positive witnesses were independently checked. We give a complete, elementary description of the integer scales preserving any admissible primitive ordered quadruple. The ten forms in the prefix comprise four infinite homothetic classes and six singleton classes. Explicit proofs identify the four infinite families, their narrower specializations and the singleton scales. Finite seed and prime-pattern studies are reported separately from global classification. The first unresolved term satisfies $130000000000 < a(41) \leq 244140625000$, with an exact upper witness but no proved index for that witness. Historical logarithmic-derivative theory is used with conservative attribution.

1 The sequence and the scope of the extension

For nonnegative integers, the arithmetic derivative is defined by $\mathrm{D}(0) = \mathrm{D}(1) = 0$ and

$$\mathrm{D}(n) = \sum_{p^e \| n} e\frac{n}{p} \qquad (n \geq 2), \tag{1}$$

where p ranges over primes. Its product rule and logarithmic derivative are classical [2, 3]. Throughout, $\mathrm{D}^2(n) = \mathrm{D}(\mathrm{D}(n))$ denotes composition, not a square. OEIS A230543 [1] lists in increasing order, with offset one, the integers $n \geq 2$ for which $\mathrm{D}(n), \mathrm{D}^2(n) > 0$ and

$$n^2 + \mathrm{D}(n)^2 + \mathrm{D}^2(n)^2 = R^2, \qquad R \in \mathbb{N}_{>0}. \tag{2}$$

We call $W(n) = (n, \mathrm{D}(n), \mathrm{D}^2(n), R)$ a witness. No primitivity, coprimality, increasing order of the three legs, or membership in a particular family is required.

The accepted OEIS baseline acquired on October 6, 2026 contains 22 terms, ending at $a(22) = 408918816$. The entry is due to Paolo P. Lava, with terms 16–18 contributed by Giovanni Resta and terms 19–22 by Ray Chandler [1]. The present study adds 18 terms relative to that dated baseline. Attempts to retrieve the current A230543 internal entry and b-file during the final checks were unsuccessful; we therefore do not assert that all 18 are unpublished in the current public entry.

The computational statement is a conditional exact extension: the accepted earlier indexing and complete negative coverage by the audited C program imply the indices displayed in Table 4. Independent trial-division checks establish membership of every positive witness, but do not independently reproduce the full negative enumeration. Section 7 separates these evidence scopes explicitly.

The structural contribution is a self-contained scale criterion for a *given* ordered primitive form, followed by its application to all ten forms observed in the prefix. Historical integer-rate theory is credited below. Ufnarovski and Åhlander's Theorem 18 [2] already solves rational logarithmic-derivative equations and gives the squarefree denominator criterion. We restate its exponent decomposition in a positive-integer form, with canonical residues and nonnegative exponent increments, without a claim of absolute novelty. The Hoepli and Sandit books of Balzarotti and Lava [4, 5] were not available in full for this study, limiting priority assessment. The overview by Tossavainen et al. [6], now consulted in full through an author-uploaded copy, surveys the history, elementary bounds and differential equations of the arithmetic derivative. No theorem is attributed to an unread book.

2 Historical tools and ordered primitive forms

For $x > 0$ put $\mathrm{L}(x) = \mathrm{D}(x)/x$. Prime exponents in a product add, so (1) gives, without a coprimality hypothesis,

$$\mathrm{D}(xy) = x\,\mathrm{D}(y) + y\,\mathrm{D}(x), \qquad \mathrm{L}(xy) = \mathrm{L}(x) + \mathrm{L}(y). \tag{3}$$

The derivative is not additive: $\mathrm{D}(2) + \mathrm{D}(3) = 2$, whereas $\mathrm{D}(5) = 1$. For a prime p, $\mathrm{D}(p) = 1$ and $\mathrm{D}^2(p) = 0$; moreover $p^2 < p^2 + 1 < (p+1)^2$. Thus primes cannot satisfy the predicate.

Kovič's Proposition 2 [3] describes the positive integer solutions of $\mathrm{D}(t) = kt$, for an integer $k \geq 1$, as $t = \prod_{j=1}^{k} p_j^{p_j}$, with repetitions permitted. The case $k = 0$ has only $t = 1$. In particular, the fixed points $\mathrm{D}(t) = t$ are precisely p^p, recorded in A051674 [7]. The 2024 survey also documents the fixed-point question in the 1950 Putnam competition and Barbeau's 1961 derivative bounds [6]. These historical scale factors will occur repeatedly.

For an integer $\lambda \geq 0$ define $T_\lambda(z) = \mathrm{D}(z) + \lambda z$ on nonnegative integers. If $\mathrm{L}(t) = \lambda$, the product rule gives

$$\mathrm{D}^j(tm) = tT_\lambda^j(m) \qquad (j \geq 0). \tag{4}$$

Indeed the claim holds at $j = 0$; if $z = T_\lambda^j(m)$ is the current integer iterate, then $\mathrm{D}(tz) = t(\mathrm{D}(z) + \lambda z)$ establishes the next step. For rational λ the same induction is valid along any chain whose displayed T_λ iterates are integers; we do not apply an integer-domain operator to an unspecified noninteger iterate. This is product-rule induction, not a binomial expansion of an additive operator.

Definition 2.1. For a witness $W = gQ$, let $g = \gcd(n, \mathrm{D}(n), \mathrm{D}^2(n), R)$ and $Q = (m, u, v, w) = W/g$. The ordered quadruple Q, with $\gcd(m, u, v, w) = 1$, is its primitive form. Its homothetic class is the set of witnesses tQ with t a positive integer.

The normalization uses all four coordinates and preserves their order. The primitive form need not itself be a derivative witness. By the product rule,

$$\lambda = \mathrm{L}(g) = \frac{u - \mathrm{D}(m)}{m}, \qquad \mathrm{D}(u) + \lambda u = v, \qquad m^2 + u^2 + v^2 = w^2. \tag{5}$$

3 A complete scale criterion for a fixed form

Theorem 3.1. *Let $Q = (m, u, v, w)$ be the primitive form of a witness and let λ be given by (5). A positive integer t produces a witness tQ if and only if $\mathrm{L}(t) = \lambda$. No noninteger positive rational multiplier produces four integer coordinates from Q.*

Proof. The equality $\mathrm{D}(tm) = tu$ is equivalent, by (3), to $\mathrm{D}(m) + m\,\mathrm{L}(t) = u$ and hence to the stated rate condition. If it holds, $\mathrm{D}(tu) = t(\mathrm{D}(u) + \lambda u) = tv$ and $(tm)^2 + (tu)^2 + (tv)^2 = (tw)^2$. All coordinates are positive. Also $tm \geq 2$: otherwise $t = m = 1$ and the first derivative equality would give $u = \mathrm{D}(1) = 0$, a contradiction. This proves necessity and sufficiency. If a/b is reduced and $(a/b)Q$ is integral, then b divides every coordinate of Q, so $b = 1$. □

The squarefree-denominator criterion and the decomposition into an integer part plus prime-denominator fractions are classical [2, Theorem 18]. The following statement explicitly restricts the exponent increments to those giving positive integer scales.

Theorem 3.2. *Suppose $\lambda = \mathrm{L}(g) = A/B$ is reduced, with $B > 0$. Then $\lambda \geq 0$ and B is squarefree. For each prime $p \mid B$ let $r_p \in \{1, \ldots, p-1\}$ be the representative satisfying*

$$r_p \equiv A(B/p)^{-1} \pmod{p}. \tag{6}$$

Put

$$t_0 = \prod_{p \mid B} p^{r_p}, \qquad \alpha_0 = \sum_{p \mid B} \frac{r_p}{p}, \qquad k = \lambda - \alpha_0. \tag{7}$$

For $B = 1$ use $t_0 = 1$ and $\alpha_0 = 0$. Then k is a nonnegative integer, and all and only the positive integer solutions of $\mathrm{L}(t) = \lambda$ are

$$t = t_0 \prod_{j=1}^{k} p_j^{p_j}, \tag{8}$$

where the primes p_j are arbitrary, may repeat, and may divide B.

Proof. Writing $g = \prod p^{e_p}$ gives $\mathrm{L}(g) = \sum e_p/p \geq 0$. This sum has a denominator dividing the squarefree product of its distinct primes, so its reduced denominator is squarefree. For any proposed $t = \prod p^{f_p}$, let M be the squarefree product of the primes dividing tB. The rate equation is equivalent to

$$\sum_{p \mid t} f_p M/p = AM/B. \tag{9}$$

If $p \mid B$ were absent from t, the left side would be zero modulo p, whereas the right side would be nonzero, since $\gcd(A, B) = 1$. Thus every prime of B divides t. Reduction modulo such a p and cancellation of M/B yield $f_p(B/p) \equiv A \pmod{p}$. Here M/B is a unit modulo p because M and B are squarefree and $p \mid B$. For a prime $p \mid t$ outside B, the right side is zero modulo p and reduction instead gives $f_p \equiv 0 \pmod{p}$.

Consequently the exponents have the form $f_p = r_p + ph_p$ for $p \mid B$, and $f_p = ph_p$ otherwise, with $h_p \geq 0$. (Absent outside primes have $h_p = 0$.) Substitution in the rate sum gives $\lambda = \alpha_0 + \sum_p h_p$. Applying this to the existing g proves that $k = \sum_p h_p$ is a nonnegative integer. Listing each prime p exactly h_p times gives (8). Conversely every product in that formula has rate $\alpha_0 + k = \lambda$. This proves both directions, including the empty product when $k = 0$. □

Corollary 3.3. *An observed homothetic class is a singleton when $k = 0$, with sole scale t_0, and infinite when $k \geq 1$.*

Proof. For $k = 0$, (8) has one value. For $k \geq 1$, fix $k-1$ parameters at 2 and vary the last through primes. The function $p \mapsto p^p$ is strictly increasing on primes, so this produces infinitely many distinct scales and first coordinates. □

The denominator condition alone is not an existence assertion for an arbitrary rational rate. Here existence is supplied by the original witness; in general the residual k must also be nonnegative. Theorem 3.2 classifies scales of an admissible form, not all admissible primitive forms of A230543. For integer rates it recovers the historical result quoted in Section 2.

4 The four infinite forms and their specializations

The four infinite forms observed are

$$\begin{aligned} Q_4 &= (2, 9, 42, 43), \\ Q_3 &= (99, 372, 1624, 1669), \\ Q_2 &= (5202, 18003, 46294, 49943), \\ Q_H &= (4802517313, 5246382600, 19023654540, 20309803913). \end{aligned}$$

Their complete first-coordinate families and scale data are shown in Table 1. Prime parameters in all formulas may coincide.

Class	First coordinate n	λ	t_0	k	Count
F_4	$2p^p q^q r^r s^s$	4	1	4	17
F_3	$99p^p q^q r^r$	3	1	3	12
F_2	$25557426p^p q^q$	37/17	4913	2	3
H	$4802517313p^p$	1	1	1	2

Table 1: The four infinite forms. Counts refer only to the 40-term prefix; the scale for F_2 is $17^3 p^p q^q$.

Proposition 4.1. *Each family in Table 1 produces witnesses for every choice of its prime parameters, and is the complete homothetic class of the corresponding ordered form.*

Proof. The exact derivative and square calculations are

$$\begin{aligned} \mathrm{D}(2) &= 1, & \mathrm{D}(9) &= 6, & 2^2 + 9^2 + 42^2 &= 43^2, \\ \mathrm{D}(99) &= 75, & \mathrm{D}(372) &= 508, & 99^2 + 372^2 + 1624^2 &= 1669^2, \\ \mathrm{D}(5202) &= 6681, & \mathrm{D}(18003) &= 7111, & 5202^2 + 18003^2 + 46294^2 &= 49943^2. \end{aligned}$$

Thus $1 + 4 \cdot 2 = 9$, $6 + 4 \cdot 9 = 42$; $75 + 3 \cdot 99 = 372$, $508 + 3 \cdot 372 = 1624$; and

$$6681 + (37/17)5202 = 18003, \qquad 7111 + (37/17)18003 = 46294.$$

For Q_H the remaining exact checks are

$$\begin{aligned} \mathrm{D}(4802517313) &= 443865287, \\ \mathrm{D}(5246382600) &= 13777271940, \\ 4802517313^2 + 5246382600^2 + 19023654540^2 &= 20309803913^2. \end{aligned}$$

For Q_H, the rate is 1, so $\mathrm{D}(m) + m = u$ and $\mathrm{D}(u) + u = v$. Each displayed quadruple has gcd one. The rates for Q_4, Q_3, Q_H are integers, so their cores are 1 and their free counts are 4, 3 and 1. For 37/17, $r_{17} = 3$, $t_0 = 17^3$, $\alpha_0 = 3/17$, and $k = 2$. Theorems 3.1 and 3.2 prove necessity and sufficiency of all four parameter formulas. Infinitude follows from Corollary 3.3. □

The integer-rate seed chains underlying narrower constructions are

λ	m	$T_\lambda(m)$	$T_\lambda^2(m)$	Root
4	2	9	42	43
3	99	372	1624	1669
2	25557426	88448739	227442422	245369959
1	102229704	353794956	909769688	981479836

For the last two rows, the first and second derivatives of the seed chain entries are, respectively,

$$\mathrm{D}(25557426) = 37333887, \quad \mathrm{D}(88448739) = 50544944,$$

$$\mathrm{D}(102229704) = 251565252, \quad \mathrm{D}(353794956) = 555974732.$$

They verify the displayed transitions directly, and each row satisfies the three-leg square identity. Formula (4) therefore also supplies direct membership proofs for these seed constructions.

Proposition 4.2. *If $g = \prod p^{e_p}$, the greatest divisor of g having an integer logarithmic derivative is*

$$E = \prod_p p^{p\lfloor e_p/p \rfloor}, \qquad \mathrm{L}(E) = \sum_p \lfloor e_p/p \rfloor. \tag{10}$$

Proof. For a divisor $d = \prod p^{f_p}$ with integer $\sum f_p/p$, multiply by the squarefree product of its primes and reduce modulo each p. This forces $p \mid f_p$. Conversely these divisibilities make the sum an integer. The largest admissible exponent not exceeding e_p is $p\lfloor e_p/p \rfloor$ independently for each prime. Their product is an admissible divisor divisible by every other admissible divisor. □

For $a(22)$, the full gcd is $78608 = 2^4 17^3$, with rate 37/17. The maximal extraction is $E = 16$, with rate 2. Dividing its witness by 16 yields the rate-2 seed row above, because $\mathrm{L}(78608/16) = 3/17$ and $(78608/16)Q_2 = 17^3 Q_2$. Dividing by 4 instead yields the rate-1 seed $4 \cdot 17^3 Q_2$. These two seeds are not themselves derivative witnesses: scaling a form of rate λ by s, with $\mathrm{L}(s) = h$, gives a seed of rate $\lambda - h$. Thus their rates are $37/17 - 3/17 = 2$ and $37/17 - 20/17 = 1$, respectively. This explains

$$F_1 = \{102229704p^p : p \text{ prime}\} \subsetneq F_2 :$$

fix one of F_2's parameters at 2. Strictness follows from the member $25557426 \cdot 3^6$, which is not divisible by 4, whereas every member of F_1 is divisible by 4. Both constructions have primitive form Q_2; they are not two independent forms.

In general, if $\mathrm{L}(s) = h$ and the relevant iterates are integral, the product rule gives

$$T_\lambda(sz) = sT_{\lambda+h}(z), \qquad T_\lambda^j(sz) = sT_{\lambda+h}^j(z).$$

This explains the rate correction when a seed is rescaled. Dividing a witness by its gcd does not preserve its derivative chain without that correction.

Fixing three parameters of F_4 at 2 gives $128p^p$; fixing two parameters of F_3 at 2 gives $1584p^p$. The former is already proved in the local comparative manuscript on A249106 [8]. Setting all parameters equal gives the diagonals $2p^{4p}$ and $99p^{3p}$. These are specializations, not additional forms. All four full families are *parametric control witnesses*: they establish positive membership and upper bounds, not consecutiveness or global coverage of the sequence.

5 Six singleton forms

Label by S_i the singleton form observed at index i. Their complete data are in Table 2; the ordered coordinates are essential even when the first is not the smallest leg.

Form	Primitive $Q = (m, u, v, w)$	λ	t_0
S_2	$(1203, 404, 408, 1333)$	0	1
S_8	$(249753, 154756, 176892, 342953)$	0	1
S_{11}	$(722718, 843189, 281066, 1145551)$	0	1
S_{28}	$(942667, 194256, 536448, 1101877)$	$2/71$	71^2
S_{36}	$(258837, 169100, 283140, 419237)$	$4/19$	19^4
S_{37}	$(43108595319, 19046592592, 38093185200, 60598798985)$	0	1

Table 2: All six singleton forms. In every row $k = 0$ and the associated term is $a(i) = t_0 m$.

Proposition 5.1. *Each form in Table 2 has exactly the indicated scale over all positive integers.*

Proof. The exact derivative and square checks are recorded below:

Form	$\mathrm{D}(m)$	$\mathrm{D}(u)$	$m^2 + u^2 + v^2 = w^2$
S_2	404	408	1776889
S_8	154756	176892	117616760209
S_{11}	843189	281066	1312287093601
S_{28}	167702	530976	1214132923129
S_{36}	114608	247540	175759662169
S_{37}	19046592592	38093185200	3672214438424437030225

The rate-zero rows have $\mathrm{D}(m) = u$, $\mathrm{D}(u) = v$, so $t_0 = 1$ and $k = 0$. For S_{28}, $u - \mathrm{D}(m) = 26554 = 2m/71$ and $v - \mathrm{D}(u) = 5472 = 2u/71$; hence $r_{71} = 2$, $\alpha_0 = 2/71$ and $k = 0$. For S_{36} the differences are $54492 = 4m/19$ and $35600 = 4u/19$, giving $r_{19} = 4$ and $k = 0$. All six quadruples have gcd one. Theorem 3.2 and Corollary 3.3 give uniqueness of the scale. □

The corresponding factorizations include

$$a(28) = 11 \cdot 17 \cdot 71^4, \qquad a(36) = 3 \cdot 19^6 \cdot 239,$$

with full gcds 5041 and 130321, respectively, and

$$a(37) = 3 \cdot 23 \cdot 31 \cdot 53 \cdot 73 \cdot 5209$$

with gcd one. Singleton here is a global statement for a fixed ordered form, not just within the scanned interval or numerical cap. It does not exclude a parametrization allowing the form itself to vary. It does disprove the claim that every term belongs to an infinite class obtained by scaling its own primitive form.

6 Exact enumeration and its correctness

The executed program uses C17, OpenMP, GNU unsigned 128-bit intermediates and GMP integers. Its certified numerical input envelope is $2 \le n \le N = 10^{16}$; the completed campaign reaches only 130000000000. The input cap is not a claim of practical complete searchability.

6.1 Predicate evaluation without family restrictions

In a block $[\ell, h]$, every integer has a residual initially equal to n and a derivative accumulator initially zero. An ordinary prime sieve provides every prime up to $\lfloor\sqrt{h}\rfloor$. For each such prime, the segmented loop visits all its multiples in the block, removes its full exponent e from the residual, and adds $e(n/p)$ to the accumulator. Afterwards a residual greater than one is prime: if it were composite, it would have a prime divisor at most $\sqrt{h}$, already removed. Adding n divided by that residual completes (1). Hence every input receives its exact first derivative.

The only early arithmetic exclusion is $\mathrm{D}(n) = 1$, the prime case proved impossible in Section 2. For a composite n, either a repeated prime contributes at least 4, or two prime factors contribute a sum at least $2 + 3$; thus this exclusion cannot remove a composite solution. No divisibility, primitivity, shape or congruence filter is used. The program factors $\mathrm{D}(n)$ exactly to compute $\mathrm{D}^2(n)$ and then forms the sum of three squares in GMP. Its `mpz_perfect_square_p` predicate and `mpz_sqrt` extraction use exact integer arithmetic [11].

Factorization first tests primality using the seven strong-probable-prime bases

$$2, 325, 9375, 28178, 450775, 9780504, 1795265022.$$

This is the deterministic test below 2^{64} stated in Forišek and Jančina's Theorem 3 [9], which credits Sinclair's base set and Izykowski's verification. The operands here are within that range. Bases are reduced modulo the operand. A zero residue is skipped; every possible exception divides one of the seven fixed bases. All 55 distinct divisors at least 2 of those bases were checked against independent trial-division primality: the implementation accepts the nine primes and rejects the 46 composites. For all other operands no base vanishes, and reduction preserves the modular-power tests in the cited theorem. This finite exceptional-case check is separate from negative coverage of the sequence. Composite splitting tries small primes through 97, then a Pollard-rho path [10], with fixed polynomials $x^2 + c$, $1 \le c \le 32$, and at most 100000 iterations per polynomial. If that path fails, exact odd trial division from 101 through $\lfloor\sqrt{x}\rfloor$ supplies a deterministic fallback. The loop guard $p \le x/p$ avoids overflow. Proper factors reduce the recursive operands; prime leaves are checked, sorted and multiplied back to the original input. Reconstruction failure is an error, not evidence of nonmembership. Rho is only a factor-finding optimization; its heuristic runtime does not enter the proof of exactness.

6.2 Numeric safety and completion semantics

For $x \ge 2$, if $\Omega(x)$ counts prime factors with multiplicity, then $\Omega(x) \le \lfloor\log_2 x\rfloor$ and

$$\mathrm{D}(x) = x \sum e_p/p \le \tfrac{x}{2}\lfloor\log_2 x\rfloor. \tag{11}$$

The corresponding real-logarithm bound is given in Ufnarovski and Åhlander's Theorem 9 [2], attributed there to Barbeau; the floor version above follows directly from $\Omega(x) \le \lfloor\log_2 x\rfloor$. At $N = 10^{16}$ it gives the conservative bounds

$$U_1 = 265000000000000000, \qquad U_2 = 7552500000000000000$$

for $\mathrm{D}(n)$ and $\mathrm{D}^2(n)$, since $\lfloor\log_2 N\rfloor = 53$ and $\lfloor\log_2 U_1\rfloor = 57$. Thus n, $\mathrm{D}(n)$ and $\mathrm{D}^2(n)$ require at most 54, 58 and 63 bits. Factorization operands are n or $\mathrm{D}(n)$, so 64 prime-factor slots suffice. Modular multiplication uses a 128-bit intermediate; even the full 64-bit operand product fits, and the actual envelope is smaller. The square sum is bounded by $N^2 + U_1^2 + U_2^2$, a 126-bit integer, and its floor square root is at most $7557154308997534291 < 2^{63}$. GMP nevertheless performs the

square computation before conversion; checked conversions enforce the stated bounds. The sieve requires primes only through 10^8 at the full input cap.

OpenMP distributes predicate evaluations with dynamic chunks of 64 and at most eight threads. Each input owns its flag, taking one of three states: negative, positive or incomplete. The loop barrier precedes serial completeness inspection and ordered positive replay. A failed factorization, out-of-range intermediate or stop request makes the block incomplete. In particular, a zero second derivative outside the already excluded prime case is not silently recorded as negative: it causes an incomplete result. For a composite n, $\mathrm{D}(n) \geq 4$, so an exact second derivative is indeed positive.

The durable frontier advances only after a complete block and successful positive replay. A segment is sealed only when its frontier equals its upper endpoint. Unsealed partial state is not full-interval evidence; interrupted blocks are repeated, not treated as intra-input checkpoints. State writes use flush, file synchronization, atomic rename and directory synchronization; cooperating commands use a run lock. These mechanisms preserve completed interval records. Their integrity checks cannot by themselves establish correctness of each negative predicate evaluation.

Proposition 6.1. *In uninterrupted successful execution within the checked numerical domain, the algorithm considers every integer in its configured interval and records exactly the witnesses satisfying* (2)*. An interval is admitted as complete only through sealed consecutive segments reaching its configured endpoint.*

Proof. Consecutive blocks partition the interval; the segmented sieve visits all inputs and computes their first derivatives as proved above. The only early exclusion is impossible for a witness. Deterministic primality, exact splitting with fallback and product reconstruction yield the second derivative, and GMP decides the square condition exactly. Any failed obligation instead prevents block completion and sealing. Serial replay records positives in increasing order. Consecutive segment bounds and a sealed endpoint therefore establish coverage, conditional on correct execution of these implemented operations. □

This is an algorithmic correctness argument supported by finite tests; it is not a machine-checked proof of the compiled implementation or a second negative enumeration.

7 The 40-term prefix and its validation

Theorem 7.1 (Computational extension, with stated dependencies)**.** *Assuming the accepted 22-term baseline and correctness of the completed C enumeration, the prefix in Table 4 is the exact indexed prefix through* 130000000000*. It has 40 terms, with* $a(40) = 129667967451$*. Its forms comprise 34 members of four infinite classes and six singleton classes. Positive membership has been checked independently; full negative coverage has not been independently enumerated a second time.*

Evidence and structural proof. The six inclusive run intervals in Table 3 are contiguous from 408918817 through 130000000000. The input count is 129591081184, comprising 18 positives and 129591081166 negatives across 22 sealed segments. The sealed parser verifies bounds, configurations, frontier completion, ordered aggregate equality and joins. Under the stated execution assumption, Proposition 6.1 excludes omitted terms in that interval. Appending the sorted positives to the accepted baseline proves the indicated indices. There is no new local negative enumeration below the baseline endpoint.

For every row, independent trial division recomputes both derivatives and an integer square root checks (2). Normalization, rate reconstruction and prime-parameter reconstruction give Table 4.

The four infinite forms are covered by Proposition 4.1, and the six remaining forms by Proposition 5.1. Their respective counts sum to $17 + 12 + 3 + 2 = 34$ and six. The complete witnesses appear in Table 5. □

Lower endpoint	Upper endpoint	Segments	Positives
408918817	711541152	1	1
711541153	10000000000	3	7
10000000001	25728640000	2	3
25728640001	47119590400	3	4
47119590401	100000000000	8	2
100000000001	130000000000	5	1

Table 3: Completed inclusive coverage after the accepted baseline.

Table 3 is the primary coverage record; the following bounded cross-checks do not replace its full negative enumeration. The independent campaign windows have a union of 10763 distinct inputs, with 11 positives and 10752 negatives. The final window is 129667966939–129667967963, containing 1025 inputs, one positive and 1024 negatives. Earlier window executions are inherited and reconciled with their original reports, not represented as fresh executions. All 40 positives were rechecked during manuscript preparation. The separate code-diagnostic union covers 33827 distinct inputs with 30 positives and 33797 negatives; it is not added to the campaign-window count.

These checks have different meanings. The independent Python oracle uses trial division and integer roots, unlike the C sieve, primality and rho route. Native replay checks state and recomputes positives, not all negatives. Archive integrity checks bind recorded files but are not mathematical certificates. The final auditor was run with ordinary Python, with `python3 -O`, and against the live retained state. Those reports agree, but they share the same audit implementation. Retained finite code tests include forced fallback and incomplete-block behavior, state corruption/recovery and parallel configurations; they support the implementation argument without certifying every input independently.

Initial family controls were known before the runs that assigned their indices. In contrast, the form Q_H first appeared in the run reaching $a(33) = 19210069252$, with parameter 2. Its parameter-3 witness was then known before the final run indexed it at 40. Thus the last witness

$$(129667967451, 141652330200, 513638672580, 548364705651)$$

confirms a previously observed form rather than introducing an eleventh.

7.1 Recorded resource use

The recorded environment is an AMD Ryzen 9 7940HS system running Linux Mint 22.3 (kernel 6.17.0-42) on x86_64 with glibc 2.39. GCC 13.3.0 compiled the program with `-O3`, `-std=c17`, `-fopenmp`, strict warning flags and `-Werror`; it links GMP 6.3.0, OpenSSL 3.0.13 and GNU OpenMP. All campaign configurations use eight threads. The sum of recorded continuation-call wall times is about 62022 seconds, or 17 hours 14 minutes. The final 100–130 billion continuation uses about 15946 seconds, or 4 hours 26 minutes, in five calls of approximately 52–54 minutes each; its maximum recorded resident set is 23536 KiB. These times exclude development, reviews and audits and are not calendar elapsed time. No full-cap timing or unmeasured parallel speedup is inferred.

The retained code, witnesses, raw interval records and independent validators are the basis of the local result. The published repository and release series in [15] supply those artifacts. Actual initial release payloads from GitHub and Zenodo were downloaded and validated with the bounded public suite; this does not repeat the complete negative scan.

8 Finite auxiliary studies and related predicates

8.1 A bounded seed rectangle

For $1 \le \lambda \le 8$ and $1 \le m \le 20000$, the seed test checks positivity of the iterates and whether $m^2 + T_\lambda(m)^2 + T_\lambda^2(m)^2$ is an integer square. The retained study covers exactly 160000 pairs, with the following 15 positives:

λ	Positive m
1	128, 864, 1584, 5832, 10692
2	32, 216, 396, 1458, 2673
3	8, 54, 99, 6250
4	2
5–8	none in this rectangle

These seeds normalize to just Q_4 and Q_3. Trial division checked the rectangle, and SymPy independently rechecked its positives; the 159985 negatives were not enumerated by a second independent implementation. Neither completeness outside the rectangle nor completeness among rational-rate primitive forms follows.

8.2 The prime pattern behind index 28

Since $187 = 11 \cdot 17$ and $\mathrm{D}(187) = 28$, the product rule, including shared-prime cases $p = 11, 17$, gives for every prime p

$$C_p = 28p + 748, \quad \mathrm{D}(187p^4) = p^3 C_p, \quad \mathrm{D}^2(187p^4) = p^2(p\,\mathrm{D}(C_p) + 3C_p). \tag{12}$$

The second equality follows by differentiating $p^3 C_p$, without any coprimality condition. The retained records cover all 168 primes $p \le 1000$ and have exactly one positive, $p = 71$, giving $a(28)$. Their read-only replay validates the stated finite screen. No assertion of uniqueness over all primes follows from it.

8.3 Actual mathematical correlations

A003415 supplies $\mathrm{D}(n)$ and A068346 supplies $\mathrm{D}(\mathrm{D}(n))$ [7]. A051674 supplies exactly the rate-one scale factors, so products of its values, together with a fixed rational-rate core, implement (8). This is a structural relation, not equality of sequence indices.

A210503 requires $n, \mathrm{D}(n)$ to be legs of a primitive Pythagorean triple. A211176 instead requires n to be the hypotenuse of a Pythagorean triple with $\mathrm{D}(n)$ a leg, without a primitivity requirement [12]. Deleting $\mathrm{D}^2(n)$ from (2) proves neither condition. The primitive-quadruple catalogues A096907–A096909 and A097263–A097265 project sorted legs or parity-selected legs, ordered by diagonal and then by the other coordinates [13]. Our Q must retain derivative order; sorting it forgets which coordinate is the original integer. A097266 counts primitive quadruples by odd diagonal, and A360946 counts quadruples by inradius [14]. Such catalogues can identify a geometric form, but do not furnish a correspondence with the increasing n indices here or settle an unsearched interval.

9 The first unresolved index and open questions

By F_4 with parameters $2, 5, 5, 5$, there is an exact witness

$$\begin{aligned} n &= 244140625000 = 2 \cdot 2^2 \cdot 5^{15}, \\ \mathrm{D}(n) &= 1098632812500, \\ \mathrm{D}^2(n) &= 5126953125000, \\ R &= 5249023437500. \end{aligned}$$

Together with completed coverage this gives, under the indexing dependencies of Theorem 7.1,

$$130000000000 < a(41) \leq 244140625000. \tag{13}$$

The upper witness is not an established value of $a(41)$ and is not in the indexed 40-term table. Negative coverage of 130000000001–244140624999, or another exact ordering proof, is still required to identify that index.

A finite parameter comparison finds this to be the smallest member above the completed endpoint among the four observed infinite classes. To make that comparison exhaustive within those classes up to the bound, it suffices to use primes $2, 3, 5, 7$: already 11^{11} exceeds the bound even before multiplication by a seed or other parameters. This does not exclude a different primitive form or an intermediate singleton.

The sequence is infinite by F_4, but infinitude of the set of *distinct primitive forms* remains open in this study. Are there infinitely many admissible singleton forms? How can admissible primitive forms be characterized before their first witness is found? The ten observed forms and two bounded screens do not justify conjectural answers to these questions. A universal cover by variable-form families would first require a precise definition of the permitted parametrizations. Saying that every term belongs to some homothetic class is merely the definition, while saying that every such class is infinite is already false.

10 Conclusions

The local computation adds 18 indexed terms to the dated accepted baseline, yielding 40 terms through 130000000000, with the negative-enumeration dependency and independent validation scopes stated explicitly. The fixed-form criterion separates complete scale classification from the unresolved classification of all primitive forms. It proves four infinite classes, accounts for their specializations, and proves six singleton classes over the full positive-integer scale domain. The next index has an exact upper witness, but remains unresolved. Finite seed and pattern studies give additional evidence without extending their bounds into global claims.

A Complete indexed and witness tables

In Table 4, g is the full quadruple gcd and the prime tuple gives the free parameters of the corresponding complete family. For S_i the tuple is empty and the sole scale is the indicated g. The F_2 scale includes its fixed core 17^3. The horizontal break after 22 distinguishes the accepted baseline from the local extension. The witness table supplies every derivative and root; both tables are part of this monolithic manuscript.

Table 4: All 40 indexed terms, forms, full gcds and free prime parameters.

i	$a(i)$	Class	g	Parameters
1	512	F_4	256	$(2,2,2,2)$
2	1203	S_2	1	$\varnothing$
3	3456	F_4	1728	$(2,2,2,3)$
4	6336	F_3	64	$(2,2,2)$
5	23328	F_4	11664	$(2,2,3,3)$
6	42768	F_3	432	$(2,2,3)$
7	157464	F_4	78732	$(2,3,3,3)$
8	249753	S_8	1	$\varnothing$
9	288684	F_3	2916	$(2,3,3)$
10	400000	F_4	200000	$(2,2,2,5)$
11	722718	S_{11}	1	$\varnothing$
12	1062882	F_4	531441	$(3,3,3,3)$
13	1948617	F_3	19683	$(3,3,3)$
14	2700000	F_4	1350000	$(2,2,3,5)$
15	4950000	F_3	50000	$(2,2,5)$
16	18225000	F_4	9112500	$(2,3,3,5)$
17	33412500	F_3	337500	$(2,3,5)$
18	105413504	F_4	52706752	$(2,2,2,7)$
19	123018750	F_4	61509375	$(3,3,3,5)$
20	225534375	F_3	2278125	$(3,3,5)$
21	312500000	F_4	156250000	$(2,2,5,5)$
22	408918816	F_2	78608	$(2,2)$
23	711541152	F_4	355770576	$(2,2,3,7)$
24	1304492112	F_3	13176688	$(2,2,7)$
25	2109375000	F_4	1054687500	$(2,3,5,5)$
26	2760202008	F_2	530604	$(2,3)$
27	3867187500	F_3	39062500	$(2,5,5)$
28	4751984347	S_{28}	5041	$\varnothing$
29	4802902776	F_4	2401451388	$(2,3,3,7)$
30	8805321756	F_3	88942644	$(2,3,7)$
31	14238281250	F_4	7119140625	$(3,3,5,5)$
32	18631363554	F_2	3581577	$(3,3)$
33	19210069252	H	4	(2)
34	26103515625	F_3	263671875	$(3,5,5)$
35	32419593738	F_4	16209796869	$(3,3,3,7)$
36	33731896677	S_{36}	130321	$\varnothing$
37	43108595319	S_{37}	1	$\varnothing$
38	59435921853	F_3	600362847	$(3,3,7)$
39	82354300000	F_4	41177150000	$(2,2,5,7)$
40	129667967451	H	27	(3)

Table 5: Complete positive witnesses for the indexed prefix.

i	n	$\mathrm{D}(n)$	$\mathrm{D}^2(n)$	R
1	512	2304	10752	11008
2	1203	404	408	1333

Table 5 (continued)

i	n	$\mathrm{D}(n)$	$\mathrm{D}^2(n)$	R
3	3456	15552	72576	74304
4	6336	23808	103936	106816
5	23328	104976	489888	501552
6	42768	160704	701568	721008
7	157464	708588	3306744	3385476
8	249753	154756	176892	342953
9	288684	1084752	4735584	4866804
10	400000	1800000	8400000	8600000
11	722718	843189	281066	1145551
12	1062882	4782969	22320522	22851963
13	1948617	7322076	31965192	32850927
14	2700000	12150000	56700000	58050000
15	4950000	18600000	81200000	83450000
16	18225000	82012500	382725000	391837500
17	33412500	125550000	548100000	563287500
18	105413504	474360768	2213683584	2266390336
19	123018750	553584375	2583393750	2644903125
20	225534375	847462500	3699675000	3802190625
21	312500000	1406250000	6562500000	6718750000
22	408918816	1415179824	3639078752	3925919344
23	711541152	3201935184	14942364192	15298134768
24	1304492112	4901727936	21398941312	21991892272
25	2109375000	9492187500	44296875000	45351562500
26	2760202008	9552463812	24563781576	26499955572
27	3867187500	14531250000	63437500000	65195312500
28	4751984347	979244496	2704234368	5554561957
29	4802902776	21613062492	100860958296	103262409684
30	8805321756	33086663568	144442853856	148445272836
31	14238281250	64072265625	299003906250	306123046875
32	18631363554	64479130731	165805525638	178874700111
33	19210069252	20985530400	76094618160	81239215652
34	26103515625	98085937500	428203125000	440068359375
35	32419593738	145888171821	680811468498	697021265367
36	33731896677	22037281100	36899087940	54635385077
37	43108595319	19046592592	38093185200	60598798985
38	59435921853	223334979084	974989263528	1002005591643
39	82354300000	370594350000	1729440300000	1770617450000
40	129667967451	141652330200	513638672580	548364705651

Data availability

The indexed prefix, documentary witness table, executed source and build instructions, sealed coverage and timing records, finite-study data and package-local validation scripts are publicly available in the GitHub repository and Zenodo release series cited in [15]. The archival identifier cited there is the concept DOI for the release series; individual deposited releases have separate version-specific DOIs. The package supplies this monolithic source and its matching PDF. Reproduction of the coverage audit requires the supplied raw records and documented validator dependencies in addition to the positive witness data; the manuscript alone does not replace those artifacts. Historical

development logs are archival evidence, not newly repeated sanitizer certification. Private reviewer conversations and third-party full texts are excluded.

AI use disclosure

OpenAI Codex and ChatGPT, Anthropic Claude, Google Gemini, xAI Grok, Alibaba Qwen and Meta AI contributed in the roles described below. The retained code reports identify ChatGPT as GPT-5.6 Sol, Claude as Opus 5.5, Gemini as 3.0 Pro, Grok as 4.7, Qwen as 3.8 and Meta AI as Muse Spark 1.1. These are reported provenance, not independently authenticated identities. Other contributions lack sufficient provider or model metadata for a reliable family attribution. Same-provider contributions are not assumed independent.

Mathematical analysis. OpenAI Codex assisted with source reading, exact-arithmetic checks, family analysis and formulation of the proofs. Historical sources and local evidence were checked separately from generated explanations.

Algorithm and code. OpenAI Codex assisted with algorithm design and implementation. The retained code-review process includes contributions from all the families named above. Frozen review and local validation records document their scope separately from the executed campaign. Reviewer execution claims were not substituted for local validation.

Manuscript. OpenAI Codex, based on GPT-6, assisted with drafting and internal six-pass checking and revision of this manuscript. Nine first-round manuscript reports were subsequently supplied by the author. Reported review systems include the six providers named above and Z.ai GLM; two reports do not establish a model/provider family. One report explicitly lacked attachment access. The reports were adjudicated against the sources and retained evidence; their execution claims were not substituted for local validation. The historical code review remains separate from manuscript review. Numerical claims depend on the retained computational artifacts and local arithmetic checks, not on conversational assertions. Carlo Corti assumes full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries and conclusions.

References

[1] P. P. Lava, OEIS A230543 (2013); description: numbers n such that n, n' and n'' are the legs of a Pythagorean quadruple, with terms from G. Resta (2013) and R. Chandler (2016–2017). https://oeis.org/A230543. Baseline acquired October 6, 2026; current endpoint not reverified.

[2] V. Ufnarovski and B. Åhlander, How to differentiate a number, *Journal of Integer Sequences* **6** (2003), Article 03.3.4. https://cs.uwaterloo.ca/journals/JIS/VOL6/Ufnarovski/ufnarovski.html.

[3] J. Kovič, The arithmetic derivative and antiderivative, *Journal of Integer Sequences* **15** (2012), Article 12.3.8. https://cs.uwaterloo.ca/journals/JIS/VOL15/Kovic/kovic4.pdf.

[4] G. Balzarotti and P. P. Lava, *La derivata aritmetica*, Hoepli, 2013. ISBN 9788820358648. https://www.hoepli.it/libro/la-derivata-aritmetica-/9788820358648.html. Publisher metadata consulted; full text not consulted.

[5] G. Balzarotti and P. P. Lava, *La derivata aritmetica*, Sandit Libri. ISBN 9788869287480. Publisher metadata consulted; publication year not independently verified and full text not consulted.

[6] T. Tossavainen, P. Haukkanen, J. K. Merikoski and M. Mattila, We can differentiate numbers, too, *The College Mathematics Journal* **55**, no. 2 (2024), 100–108. DOI: 10.1080/07468342.2023.2268494. https://www.researchgate.net/publication/375299116_We_Can_Differentiate_Numbers_Too. Full author-uploaded text consulted October 9, 2026.

[7] N. J. A. Sloane and R. K. Guy, OEIS A003415, arithmetic derivative; R. Zumkeller, OEIS A068346, second arithmetic derivative; A. Auel, OEIS A051674, p^p for prime p, with T. D. Noe's fixed-point interpretation (2004). https://oeis.org/A003415; https://oeis.org/A068346; https://oeis.org/A051674. Definitions consulted October 9, 2026.

[8] C. Corti, *An Exact Extension and an Infinite Family for OEIS A249106*, local manuscript, version 4 (2026), section on A230543. Public availability not verified.

[9] M. Forišek and J. Jančina, Fast primality testing for integers that fit into a machine word, in *Student Research Forum Papers and Posters at SOFSEM 2015*, CEUR Workshop Proceedings **1326** (2015), 20–30. https://ceur-ws.org/Vol-1326/020-Forisek.pdf.

[10] J. M. Pollard, A Monte Carlo method for factorization, *BIT* **15** (1975), 331–334. DOI: 10.1007/BF01933667. https://pages.cs.wisc.edu/~cs812-1/pollardrho.pdf.

[11] GNU MP project, GNU MP 6.3.0 manual, Section 5.8, Root extraction functions. https://gmplib.org/manual/Integer-Roots. Consulted October 9, 2026.

[12] P. P. Lava, OEIS A210503, arithmetic-derivative primitive Pythagorean triples, and OEIS A211176, arithmetic-derivative hypotenuse condition (2013). https://oeis.org/A210503; https://oeis.org/A211176. Definitions consulted October 9, 2026.

[13] R. Chandler, OEIS primitive Pythagorean quadruple projections A096907–A096909 and A097263–A097265 (2004). https://oeis.org/A096907; https://oeis.org/A096908; https://oeis.org/A096909; https://oeis.org/A097263; https://oeis.org/A097264; https://oeis.org/A097265. Definitions consulted October 9, 2026.

[14] R. Chandler, OEIS A097266, Number of primitive Pythagorean quadruples with diagonal $2n+1$; M.-Á. Pérez García-Ortega, OEIS A360946, Number of Pythagorean quadruples with inradius n (2023). https://oeis.org/A097266; https://oeis.org/A360946. Definitions consulted October 9, 2026.

[15] C. Corti, *An Exact Local Extension and Homothetic Classes for OEIS A230543*, GitHub repository and Zenodo archive, 2026. https://github.com/carcorti/A230543; https://doi.org/10.5281/zenodo.23311392. The DOI identifies the archived release series; each deposited release has a separate version-specific DOI.